Construction Calculator Pro: Package Index

This package turns the workbook and guide into one solution for planning, pricing and paying for a build or renovation in India. Start with folder 2, then work through folder 1 in order.

1. Core product

- **1_Core/01_Main_Guide.md:** eight modules that follow the order of the workbook, with worked checks and a list of what the file does not calculate.

2. Quick start

- **2_Quick_Start/00_Start_Here.md:** the first 20 minutes, the route to a costed estimate, and what to do before paying anyone.

3. Implementation

- **3_Implementation/02_Action_Plan.md:** a four-week plan, with an output for each week.
- **3_Implementation/03_Worksheets.md:** printable forms for measuring, paint, outside walls and roof, quotes, bills and payments.
- **3_Implementation/04_Checklists.md:** before you start, stage checks, before each payment, and before handover.
- **3_Implementation/05_Exercises.md:** five exercises, with answers.

4. Tools

- **4_Tools/06_Templates.md:** rate request, change order, stage payment record and approval fee record.
- **4_Tools/07_Scripts.md:** short messages for quotes, bills, change orders and holding payments.
- **4_Tools/08_Trackers.md:** which workbook tracker to update, when, and who checks it.
- **4_Tools/09_Cheat_Sheet.md:** units, formulas and a check before you trust a figure.

5. Bonuses

- **Bonus 1:** Contractor Quote and Bill Check
- **Bonus 2:** Payment Schedule Guide
- **Bonus 3:** Change Order Control
- **Bonus 4:** Specification and Brand Sheet: fix brands and grades so every quote prices the same thing
- **Bonus 5:** Material Quality Checks on Site: field checks for each material on delivery, with a receipt log

- **Bonus 6:** Contractor Hiring Kit: questions, site visits, reference checks and a scoring sheet
- **Bonus 7:** Snag List and Handover Check: room-by-room defect checks before the final payment

6. Workbook

- **Construction_Calculator_Pro_INDIA_v2.xlsx:** the blank calculator. Use this for your own project.
- **Construction_Calculator_Pro_EXAMPLE_filled.xlsx:** a 30 × 50 ft new house with placeholder figures.
- **Construction_Calculator_Pro_EXAMPLE_renovation.xlsx:** a kitchen, bathroom and bedroom renovation with placeholder figures.

Read before you rely on any figure

- The Terms & Disclaimer tab in the workbook.
- The Assumptions tab. Most factors have not yet been checked by an engineer.
- The example files. Their totals are placeholders, not benchmarks.

Construction Calculator Pro. General guidance only. Your engineer, contractor, lawyer and local authority decide what applies to your project.

Start Here: Plan, Price and Pay for Your Build

What this package does: it helps you turn your plan into quantities, a costed estimate, a purchase list and a payment plan, so you can compare quotes and check bills before you pay.

What it does not do: it is not a structural design, and it does not replace an engineer, architect, quantity surveyor or lawyer. Read the Terms & Disclaimer tab in the workbook before you rely on any figure.

Before you start: open the example file first if you want to see a finished project. Its figures are placeholders. Then use the blank file for your own project.

The fastest route to a first result (about 20 minutes)

1. Open the workbook. On **Quick Start**, answer the six questions: project name, client, built-up area, budget, city, and Feet or Metres. Choose your units before you measure anything.
2. On **Room Area**, enter each room once (up to 12 rooms).
3. On **Paint & Tiles**, enter each room's wall height and its doors and windows.
4. On **Exterior & Roof**, enter the outside wall length and height, and the flat roof area, if you have them.
5. On **Earthwork**, enter the trenches and pits.
6. On **Cost Estimate**, look at the quantities that have appeared. You will not see money figures yet.

The route to a costed estimate (1 to 2 days)

7. Get at least two quotes for each major material (cement, steel, bricks, tiles, paint). Follow Module 4 of the Main Guide.
8. Enter your rates on the **Rates** tab, with the source and date. Until you do, every cost shows zero.
9. Count plumbing and electrical points on the **Services** tab.
10. Check the **Cost Estimate** total, the **Stage Budget** and the warnings.

Before you pay anyone

11. Use the **Order List** to plan purchases and order-by dates.
12. Use the **Bill Check** tab to compare any contractor's quantities with yours.
13. Use the **Payments** tab to record each payment and change order, and the **Permits** tab to record approval fees.

Read these first

- **Assumptions** tab: every factor the calculations use, and whether it has been checked.
- **Terms & Disclaimer** tab: what this file is and is not.
- **Example file (new build):** a 30 × 50 ft house with placeholder figures, to show how the file works.
- **Example file (renovation):** a kitchen, bathroom and bedroom with no structure, to show a smaller job.
- Do not use the example totals as a benchmark for your own project.

Construction Calculator Pro. Estimates only. Check every figure with your engineer, your contractor and your local authority. Read the Terms & Disclaimer tab in the workbook before relying on any figure.

Plan, Price and Pay for Your Build: The Main Guide

This guide follows the order of the workbook. Each module matches one or more tabs. Work through the modules in order the first time, then come back to any module when you need it.

Before you start, read the Terms & Disclaimer tab in the workbook.

Module 1: Set up your project

Time: 10 to 15 minutes. **You'll have:** one set of project settings that every calculator uses.

Enter the project name, client, city, project type (new build, renovation or interior only), units, built-up area, target budget, contingency and GST on the Quick Start and Setup tabs.

- **Units:** choose Feet or Metres before you measure anything. Changing the units does not convert numbers you have already typed. Re-enter them after switching.
- **Built-up area:** the total floor area you are building, including walls.
- **Target budget:** the most you plan to spend. Write down the figure you won't go above.
- **Contingency:** an allowance for unexpected costs. The 10% default is a setting you choose, not a rule.
- **GST:** 0% is a placeholder. Confirm the rate for your contract with your accountant before you use any total that includes GST.

Module 2: Measure once

Time: 30 to 60 minutes, plus measuring. **You'll have:** a measured list of rooms, walls, openings and roof area, with each figure entered once.

Measure the largest room twice. If the readings differ, measure a third time and use the middle figure.

- **Room Area:** room length, width and count. For identical rooms, use the count column.
- **Paint & Tiles:** each room's wall height, and the number of doors and windows in it. Door and window sizes are set once at the top of the tab.
- **Exterior & Roof:** the outside wall length (perimeter), the outside wall height, the total area of doors and windows on outside walls, and the plan area of a flat roof.
- **Bricks & Blocks:** wall lengths and heights. Openings are linked from Paint & Tiles: windows are taken off external walls, and doors off internal walls. If a door is in an outside wall, move it by hand.

Check: the Room Area total should be below the built-up area, because walls take up space. In the example, about 1,000 sq ft of rooms sits inside 1,500 sq ft built-up. A much larger gap suggests a missing room.

Module 3: Calculate quantities

Time: 45 to 60 minutes. **You'll have:** a quantity for each material, with wastage included.

- **Concrete:** wet volume × 1.54 gives dry volume. Cement bags = dry volume × cement parts ÷ total parts ÷ 1.25 cu ft per bag, rounded up.
- **Bricks:** net wall area × thickness gives volume in cubic metres. Bricks = volume × 500 × 1.05, rounded up. AAC blocks are a separate option.
- **Steel:** weight per metre = diameter² ÷ 162 kg. Bar weight = weight per metre × length.
- **Interior paint:** litres = net wall area × coats ÷ coverage × 1.10, rounded up. Ceilings are not painted in the calculation.
- **External paint:** the same method, using the outside wall area on the Exterior & Roof tab.
- **Roof waterproofing:** flat roof area × the roof rate on the Rates tab. The rate should cover material and labour. Pitched roofs are not covered.
- **Tiles:** tiles = floor area × 1.10 ÷ area of one tile, rounded up.
- **Plaster:** cement and sand for walls and ceilings, plus plaster labour at the rate on the Rates tab. External wall faces are not included.

Worked check: paint for one room. A 12 ft × 10 ft room, 10 ft high, with one door (3 × 7 ft) and one window (4 × 4 ft), two coats at 100 sq ft per litre. - Walls: 2 × (12 + 10) × 10 = 440 sq ft - Less openings: 440 − 21 − 16 = 403 sq ft - Paint: 403 × 2 ÷ 100 × 1.10 = 8.9, so **9 litres**

Compare your hand figure with the file. If they differ, check the openings first.

All the factors used here are listed on the **Assumptions** tab. Most have not yet been checked by an engineer. Confirm the ones you rely on.

Module 4: Set your rates

Time: 1 to 2 hours, spread over a few days. **You'll have:** a dated rate for every material and labour item, each with its source.

Enter rates on the Rates tab. - Materials: cement (per bag), M-sand and aggregate (per cu ft), bricks and AAC blocks (per piece), steel (per kg), paint (per litre), exterior paint (per litre), tiles (per sq ft). - Labour: concrete (per cu ft), masonry (per sq ft), plaster (per sq ft of plaster), interior painting (per sq ft of wall), tile laying (per sq ft of floor). - Other: excavation (per cu ft), roof waterproofing with labour (per sq ft of roof).

Rules for rates: - Get at least two quotes for cement, steel, bricks and tiles. Use the Rate Quotes tab to compare them. - Use the unit the file uses. If a supplier quotes sand per tonne, convert it to cubic feet using the density the supplier gives. Do not guess the density. - Ask whether each quote includes delivery, loading and GST. Write the answer in the Source column.

The rates drive every material and labour cost in the calculators. The target budget and contingency on Setup are separate money figures you also enter.

Module 5: Budget by stage

Time: 30 to 45 minutes. **You'll have:** a target amount for each build stage, and a check that stage costs fit inside your budget.

Check that every cost line on the Cost Estimate has a stage. Set a target share for each stage on the Stage Budget tab. The shares shown are examples only. Set your own to match your plan. Target shares should add up to 100%.

Stage costs are before contingency and GST. The Cost Estimate total includes both.

Module 6: Order materials

Time: 30 to 45 minutes. **You'll have:** a purchase list with quantities, wastage, costs, suppliers and order-by dates.

The Order List takes its quantities from the calculators. Wastage defaults: sand 10%, aggregate and steel 5%, and cement, bricks and tiles 0% (bricks and tiles already include breakage). Confirm these with your supplier.

Enter a supplier and an order-by date for each line. Work backwards from the stage that needs the material, and allow for delivery time. The order-by dates are typed in by hand; the file does not work them out from lead times.

Avoid ordering everything on one day. Cement and sand are usually needed closer to the pour date than bricks.

Module 7: Track payments, approvals and changes

Time: 30 minutes to set up, then 5 minutes each payment. **You'll have:** a record of every payment, approval fee and agreed change, and the balance still to pay.

Record each payment on the day you make it, with the payment type: deposit, stage payment, final payment, retention or other. Record change orders as positive amounts for extra work and negative amounts for savings. Set "Approved?" to Yes only when both sides have agreed in writing.

Record approval and professional fees on the Permits tab, one fee per approval. Their total flows to the Cost Estimate.

- **Revised forecast** = the estimated total plus approved changes.
- **Balance still to pay** = revised forecast − total paid.
- A negative balance means you have paid more than the revised forecast. Check for a duplicate payment, a wrong change order, or a payment made before a change was approved.

- If approved changes are paid out of contingency, do not add them to the forecast as well. Reduce the contingency line instead, so the cost is counted once.

The payments table has room for 15 payments and 10 change orders.

Module 8: Check a quote or bill (optional)

Time: 30 minutes per quote or bill. **You'll have:** a comparison of each claimed quantity with your calculation.

Enter each quantity the contractor claims on the Bill Check tab. The file flags any claim more than 5% above or below your calculation. The 5% tolerance can be changed.

- **Claim is higher:** ask for the measurement sheet behind it. A higher claim is not always wrong. The contractor may have included wastage, or measured a room you left out.
- **Claim is lower:** check that the scope is complete.
- Compare like with like: the same units, the same rooms, and the same wastage allowance.

The Bill Check compares quantities only. It does not check rates or amounts.

What the file does not calculate

The workbook shows a total only for what it calculates. Add these yourself on the Cost Estimate, with a stage for each:

- Kitchen counter and cabinets, sanitaryware and taps, doors and windows, and their fittings
- Pitched roofs, and external plaster
- General labour not covered by the calculators, transport and unloading
- Scaffolding, cleaning, and crack repair

Stage-by-stage notes

The Stage Checklist tab and the Checklists in folder 3 list checks for each build stage. These are general points. Your engineer or site supervisor decides the actual requirements. Keep photos and signed records of each check.

Construction Calculator Pro. Estimates only. Check every figure with your engineer, your contractor and your local authority. Read the Terms & Disclaimer tab in the workbook before relying on any figure.

Action Plan: From Idea to Signed Estimate (4 weeks)

Work through the weeks in order. Each step says what to produce, so you know when it is done. A renovation can skip the structure steps in Week 2.

Week 1: Set the target

- [ ] Write the budget you can afford, and the figure you will not go above. (Module 1)
- [ ] Decide the built-up area, city and units. (Module 1)
- [ ] Get your plan. Use the architect's or your own sketch. (Module 2)
- [ ] Check the approvals list with your local authority. (Permits tab)
- [ ] **Output:** Quick Start filled in, and the approvals list started.

Week 2: Measure and quantify

- [ ] Measure every room twice. Enter each room once on Room Area. (Module 2)
- [ ] Enter wall heights, doors and windows on Paint & Tiles. (Module 2)
- [ ] Enter the outside wall length, height and openings, and the flat roof area, on Exterior & Roof. (Module 2)
- [ ] Enter trenches and pits on Earthwork. (Module 3)
- [ ] Enter concrete items, steel bars and wall groups. (Module 3)
- [ ] Enter plumbing and electrical points on Services. (Module 3)
- [ ] Check the room total against the built-up area. (Module 2)
- [ ] **Output:** quantities for every material, with no blank input cells in the sections you need.

Week 3: Collect rates and build the estimate

- [ ] Get at least two quotes for cement, steel, bricks, tiles and paint. (Module 4)
- [ ] Get labour rates for plaster, painting and tile laying, and a roof waterproofing rate that includes labour. (Module 4)
- [ ] Enter rates on the Rates tab, with the source and date. (Module 4)
- [ ] Use Rate Quotes to compare the two suppliers on matching delivery terms. (Module 4)
- [ ] Enter approval and professional fees on Permits. (Module 7)
- [ ] Add the costs the calculators don't cover on Cost Estimate, each with a stage. (Module 3)
- [ ] Check the Cost Estimate total, the Stage Budget and the warnings. (Module 5)
- [ ] Compare the total with your target budget. Decide what to change. (Module 5)
- [ ] **Output:** a costed estimate you can defend.

Week 4: Plan the buying and the payments

- [ ] Set stage targets on the Stage Budget tab, totalling 100%. (Module 5)
- [ ] Enter the Order List suppliers and order-by dates. (Module 6)
- [ ] Get written terms from each contractor: scope, payment stages, change-order process. (Bonuses 2 and 3)
- [ ] Record your first payment, even a small deposit, on Payments. (Module 7)
- [ ] **Output:** a purchase list, a payment plan and a record of approvals.

Ongoing

- [ ] Check every bill against your quantities. (Module 8)
- [ ] Record every change order, and do not start extra work until it is approved in writing. (Bonus 3)
- [ ] Update rates each time you get a new quote.
- [ ] Complete the stage checks before each payment. (Checklists)

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Worksheets

Print these pages, or copy them into a notebook. Fill them in before you enter numbers in the workbook, then check the workbook against them.

Worksheet A: Room measurement log (Module 2)

Room	Length (1st / 2nd)	Width (1st / 2nd)	Height	Doors (count and size)	Windows (count and size)	Final length	Final width
	/	/					
	/	/					
	/	/					

Measure twice. If the two readings differ by more than a few centimetres, measure again.

Worksheet B: Outside walls and roof (Exterior & Roof)

Item	Figure	Unit
Outside wall length (all sides)		
Outside wall height		
Area of doors and windows on outside walls		
Outside wall area (length × height − openings)		
Flat roof plan area		

Worksheet C: Paint by hand (Module 3)

Step	Figure
Walls: 2 × (length + width) × height	sq ft
Less openings	sq ft
Net wall area	sq ft
Coats	
Paint (net area × coats ÷ coverage × 1.10)	litres

Worksheet D: Quote comparison (Module 4)

Item	Supplier A rate and unit	Delivered? (Y/N)	Supplier B rate and unit	Delivered? (Y/N)	Same unit?	Lower rate, same terms
Cement						
Steel						
Bricks						
Tiles						
Paint						

Worksheet E: Bill check (Module 8)

Item	Your figure	Contractor's claim	Difference (%)	Above 5%?	Question to ask

Worksheet F: Payment balance (Module 7)

Line	Amount (₹)
Estimated total	
Approved changes (only those signed by both sides)	
Revised forecast	
Total paid to date	
Balance still to pay	
Balance is negative? If yes, why?	

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Checklists

Before you start building

- [ ] Budget and maximum written down (Module 1)
- [ ] Units chosen before measuring (Module 1)
- [ ] Rooms measured twice and entered once (Module 2)
- [ ] Rates entered with source and date for every item (Module 4)
- [ ] Assumptions tab reviewed, and the items you rely on are confirmed (Assumptions tab)
- [ ] Approvals list checked with your local authority, and fees entered on Permits (Permits tab)
- [ ] Written contract or quote covers scope, payment stages and change orders (Bonuses 2 and 3)

Stage checks (before moving to the next stage)

Site preparation and foundation - [ ] Soil test report received - [ ] Layout marked and checked against the approved plan - [ ] Foundation depth and width inspected before concrete is poured

Structure (concrete and steel) - [ ] Steel bar size, spacing and cover match the drawings before concrete - [ ] Concrete grade as specified; test cubes taken - [ ] Formwork checked for level and alignment - [ ] Curing started and recorded - [ ] Flat roof waterproofed after the slab is cured, and tested by ponding water before any finish is laid

Masonry (walls) - [ ] Walls checked for alignment and vertical level - [ ] Mortar mix as specified - [ ] Lintels or bands in place above openings - [ ] Plaster applied to the thickness specified, and cured

Flooring - [ ] Floor surface level and dry - [ ] Tile layout planned before fixing - [ ] Tiles from the same batch - [ ] Joints checked after fixing

Painting (interior and exterior) - [ ] Walls fully dry and plastered - [ ] Primer applied - [ ] Number of coats as specified - [ ] Paint batch numbers recorded

Doors, windows and fittings - [ ] Frames fixed plumb and level - [ ] Hardware and glass match the order - [ ] Gaps around frames sealed

Plumbing and electrical - [ ] Water lines pressure-tested - [ ] Electrical circuits and earthing tested - [ ] Concealed work photographed before walls are closed

Kitchen and bathrooms - [ ] Waterproofing tested (ponding test) before tiling - [ ] Fittings match the order - [ ] Drainage tested to flow freely

Approvals, fees and other - [ ] Building approval in hand before work that needs it starts - [ ] Completion certificate requested - [ ] Final payment made only after final inspection

Before each payment

- [ ] Stage checks for the work being paid for are complete
- [ ] Quantities match the Bill Check (or differences are explained)
- [ ] Any extra work is in a written, approved change order
- [ ] Payment recorded on the Payments tab
- [ ] Receipt or transfer record kept

Before handover

- [ ] Final inspection done
- [ ] Completion certificate received
- [ ] Final payment made after inspection
- [ ] Warranties and manuals collected

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Exercises

Do each exercise before you enter your own numbers. Exercises 2 and 4 use fixed figures so you can check your answer. Exercises 3 and 5 use illustrative figures, not the example file.

Exercise 1: Measure a room twice (Module 2)

Measure one room of your own twice, with the same tape. Write both readings on Worksheet A.

Check: is the difference under a few centimetres? If not, measure a third time and use the middle figure.

Exercise 2: Paint one room by hand (Module 3)

A 12 × 10 ft room, 10 ft high, one door (3 × 7 ft), one window (4 × 4 ft), two coats, 100 sq ft per litre.

Answer: - Walls: 2 × (12 + 10) × 10 = 440 sq ft - Openings: 21 + 16 = 37 sq ft - Net wall area: 440 − 37 = 403 sq ft - Paint: 403 × 2 ÷ 100 × 1.10 = 8.9, so **9 litres**

If your answer differs, check the openings first.

Exercise 3: Compare two quotes (Module 4)

Illustrative figures. Cement quoted per 50 kg bag.

Supplier	Rate per bag (₹)	Delivered?	Delivery charge per bag (₹)	Cost per bag, delivered (₹)
A	400	No	15	415
B	410	Yes	0	410

Answer: B is cheaper by ₹5 per bag once delivery is included, even though its base rate is higher. Compare only quotes that include the same delivery and loading terms.

Repeat this with two real quotes for cement and steel on Worksheet D.

Exercise 4: Find the gap in a bill (Module 8)

Illustrative figures. The contractor bills 25 cu ft of concrete for a slab. Your calculation gives 22.0 cu ft.

Answer: the difference is (25 − 22.0) ÷ 22.0 = 13.6%. That is above the 5% tolerance, so ask for the measurement sheet.

Possible causes to check: wastage added twice, a wrong slab thickness, a different unit, or a room counted twice.

Exercise 5: Balance a payment (Module 7)

Illustrative figures.

Line	Amount (₹)
Estimated total	10,00,000
Approved changes (both sides signed)	50,000
Revised forecast	10,50,000
Total paid to date	3,00,000
Balance still to pay	7,50,000

Answer: the balance is positive, so it is safe to pay the next stage, if the stage checks are complete. If the balance were negative, find out why before you pay anything else.

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Templates

Copy these into a document or a notebook. Keep a signed copy of each one.

Template 1: Rate request (send to each supplier)

> Hello [name], I am planning a [project type] at [address/area] and would like a quote for the following, delivered to the site:
>
> - [Material 1]: [quantity] [unit]
> - [Material 2]: [quantity] [unit]
>
> Please quote per [unit], and tell me whether the price includes delivery, unloading and GST. Please also send the quote date and how long it is valid for. Thank you.

Template 2: Change order

Field	Entry
Change order number	
Date	
Description of the change	
Reason for the change	
Quantity and unit	
Rate and amount (₹)	
Saving elsewhere, if any (₹)	
Effect on the programme (days)	
Approved by owner (name, signature, date)	
Approved by contractor (name, signature, date)	

Do not start the work until both signatures are on the form.

Template 3: Stage payment record

Field	Entry
Payment number	
Stage paid for	
Stage checks complete? (Yes/No)	

Field	Entry
Quantities checked against Bill Check? (Yes/No)	
Amount (₹)	
Date paid	
Method (cheque, transfer, cash with receipt)	
Receipt or transfer reference	

Template 4: Approval fee record

Approval	Who issues it	Fee (₹)	Applied on	Approved on	Reference number

Record the same details on the Permits tab, so the total flows to the Cost Estimate.

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Scripts

These are short messages you can send by phone, WhatsApp or email. Change the details to suit your project. They are written to keep the record clear, not to pressure anyone.

Script 1: Asking for a quote with the same scope

> Hello, I am comparing quotes for [item]. Could you quote for [quantity and unit], delivered to [site], and tell me whether the price includes delivery, loading and GST? I will compare it with two other quotes.

Script 2: Asking for the measurement sheet

> Thank you for the bill. For [item], it shows [claimed quantity]. My calculation is [your quantity]. Could you send the measurement sheet for this item, so I can see which area it covers?

Script 3: Asking for a change to be written down

> Before we start the extra work for [description], please could you send a written change order with the quantity, rate and amount? I will confirm it in writing once I have checked it.

Script 4: Holding a payment until checks are done

> I am ready to pay the [stage name] amount. Before I do, I need to complete the stage checks, which are [list]. I will pay within [number] days of those checks being complete.

Script 5: Asking about the next stage

> Thank you for the update. Could you tell me what will be finished for the next stage, when you expect it, and what you need from me to start it?

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Trackers

The workbook has trackers that record what has happened. Update them as you go. The table shows which tab to use, when to update it, and who should check it.

Tracker (workbook tab)	What it records	When to update	Who checks it
Task Tracker	Tasks by stage and room, with the person responsible, due date and overdue flag	Weekly, and whenever a task is done	Owner
Stage Checklist	Checks to complete before each stage is closed	Before each payment and each stage	Owner, with engineer or supervisor
Order List	Materials to buy, with wastage, supplier and order-by date	When the quantities change, and when you place each order	Owner
Permits	Approvals, their status, the applied and approved dates, and the fee for each	When an application is made, approved or rejected	Owner
Payments	Each payment, its type, change orders, and the balance still to pay	On the day of each payment, and when a change is approved	Owner
Bill Check	Contractor quantities against your calculation, flagged above the tolerance	When each bill arrives	Owner
Rate Quotes	Two supplier quotes compared on matching terms	When each quote arrives	Owner
Timeline	Tasks with dates, progress and days left	Weekly	Owner
Quotation	A printable quotation for a client	When a quotation is issued	Issuer

Keep a simple routine

- **Daily or weekly:** Task Tracker and Timeline
- **Each payment:** Payments, Stage Checklist and Bill Check
- **Each approval or fee:** Permits
- **Each order:** Order List and Rate Quotes

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Cheat Sheet: Units and Formulas

Conversions

- 1 m = 3.28084 ft
- 1 sq m = 10.7639 sq ft
- 1 cu m = 35.3147 cu ft
- 1 ft = 304.8 mm

Quantities

Item	Formula	Notes
Dry concrete	wet volume × 1.54	Common practice. Engineer to confirm.
Cement bags (50 kg)	dry volume × cement parts ÷ total parts ÷ 1.25	1.25 cu ft per bag
Bricks	volume (m^3) × 500 × 1.05	500 per cu m is an approximation
Steel weight	diameter² ÷ 162 × length (m) × count	kg per bar
Interior paint (litres)	net wall area × coats ÷ coverage × 1.10	Walls only, no ceilings
External paint (litres)	outside wall area × coats ÷ coverage × 1.10	Exterior & Roof tab
Roof waterproofing	flat roof area × roof rate	Rate covers material and labour
Tiles	floor area × 1.10 ÷ area of one tile	Tile area in sq ft
Plaster	area × thickness ÷ 304.8 × 1.27 × 1.10	Split 1:6 cement to sand
Excavation	length × width × depth × count, then + 10%	Cu ft

Money

- Cost Estimate total = subtotal + contingency + GST
- Stage target (₹) = target share × target budget
- Revised forecast = estimated total + approved changes (or the contingency reduction, if changes are paid from contingency)
- Balance = revised forecast − total paid

Before you trust a figure

- Have you entered every rate? Blank rates show zero.
- Are the units right? Changing units does not convert numbers already typed.

- Is the assumption on the Assumptions tab confirmed?
- Does every cost line you added have a stage?
- Are the approval fees entered on Permits?

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Bonus 1: Contractor Quote and Bill Check

DRAFT. Check every point against local practice and your contract before sale.

What it solves

You can't tell whether a quote or running bill is fair.

How to use it

1. Get the contractor's quote or bill with its quantities written out.
2. Enter each claimed quantity on the Bill Check tab.
3. For every item marked higher than your calculation, ask for the measurement sheet.
4. Check the common causes below before you conclude the bill is wrong.

Common causes of a higher claim

- Wastage added on top of your figure, when your Order List already includes it.
- A different unit (for example, cubic metres where you used cubic feet).
- A room counted twice, or an opening not deducted.
- Work measured to a different thickness or height than the drawings.

Questions to ask

- "Can you send the measurement sheet for this item?"
- "Which room or wall does this figure cover?"
- "Is the wastage included, and at what percentage?"

What it does not check

Rates and amounts. Compare the rates separately, using Module 4.

Construction Calculator Pro. Estimates only. Check every figure with your engineer, your contractor and your local authority. Read the Terms & Disclaimer tab in the workbook before relying on any figure.

Bonus 2: Payment Schedule Guide

DRAFT. This is general guidance, not legal advice. Check the terms with a lawyer before you use them.

What it solves

Paying ahead of the work, which leaves you with no leverage if the contractor stops.

What to agree before the first payment

- The scope, in writing, with quantities or a reference to the drawings.
- The payment stages, and the work that must be complete before each one.
- How the final payment is handled, including any amount held back until after inspection.
- How change orders are approved (see Bonus 3).

Warning signs

- Being asked for a large payment before any work has started, other than a modest deposit.
- Payments requested for stages that have not been reached.
- Being asked to pay in cash without a receipt.

Before each payment

- The stage checks for that work are complete (see the Checklists).
- The quantities match the Bill Check, or the differences are explained.
- Any extra work is in an approved, written change order.
- The payment is recorded on the Payments tab, with a receipt or transfer record.

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Bonus 3: Change Order Control

DRAFT. Check the wording with a lawyer before you use it in a contract.

What it solves

Extra work agreed verbally, which later appears on the bill as an unexpected cost.

The rule

Do not start extra work until the change is written down, priced and approved by both sides.

A simple change order

Field	Entry
Date	
Description of the change	
Reason for the change	
Quantity and unit	
Rate and amount (₹)	
Effect on the programme (days)	
Approved by owner (name and date)	
Approved by contractor (name and date)	

Pricing a change before you approve it

- Use the rates on your Rates tab, or get a quote for the new work.
- Check whether the change saves money elsewhere, and record a negative change order if so.
- Check that the change does not affect the contingency you planned to keep.

Recording it

Enter the change on the Payments tab under Change orders. Set Approved? to Yes only after both signatures. Pending changes are not added to the revised forecast.

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Bonus: Specification and Brand Sheet

Fill this in before you ask for quotes, so every contractor prices the same thing.

Why this matters

Two quotes for "flooring" can differ by lakhs, and both can be honest. One may price a ₹45 tile and the other a ₹110 tile. One may include skirting and the other may not. Until the specification is fixed, you are comparing different houses, not different prices.

This sheet fixes the brand, grade, size and quantity basis of every major item. Attach it to every quote request. Then the only thing that changes between quotes is the price.

How to use it

1. **Fill in the structure section with your engineer.** The concrete grade, steel grade and cement type come from the structural design, not from your choice.
2. **Fill in the finishes yourself.** Visit one or two showrooms, choose, and write the brand, range and code.
3. **Where you have not chosen yet, set an allowance.** Write a rate per unit (for example, "floor tile, supply only: ₹70 per sq ft"). The contractor prices the work using that allowance. When you choose the final item, the difference is added or taken off by a written change order.
4. **Avoid "or equivalent" on its own.** If you must use it, write "or equivalent, approved in writing by the owner before purchase".
5. **Say who supplies each item.** "Owner supplies" means you buy it and the contractor only fixes it. "Contractor supplies" means the price includes the material.
6. **Send the same sheet to every contractor**, and ask them to quote against each line.

Who supplies: the three options

Option	What it means	When it suits you
Contractor supplies and fixes	Material and labour in one price	You want one point of responsibility
Owner supplies, contractor fixes	You buy, the contractor installs	You want control of brand and price, and can manage deliveries
Allowance (provisional sum)	A fixed amount per unit in the quote, adjusted later	You haven't chosen yet, but need a comparable quote now

Section 1: Structure (from your engineer)

Item	Specification	Brand / source	Who supplies	Notes
Cement	Type and grade (for example, OPC 53 or PPC)			Engineer to confirm
Steel (TMT bars)	Grade (for example, Fe 500 or Fe 500D) and sizes			Engineer to confirm
Concrete	Grade for footings, columns, beams and slab	Site mix or ready-mix		Engineer to confirm
Sand	M-sand or river sand; plastering or concrete grade			
Coarse aggregate	Size (for example, 20 mm and 12 mm)			
Walling	Red brick or AAC block; size and thickness			
Waterproofing (roof)	System and product			
Waterproofing (bathrooms)	System and product			
Anti-termite treatment	Product and stage			

Section 2: Flooring and wall tiles

Area	Tile type	Size	Brand and range	Code	Supply rate or allowance (₹ per sq ft)	Skirting included?	Who supplies
Living and dining							
Bedrooms							
Kitchen floor							
Kitchen wall (dado)							
Bathroom floor (anti-skid)							
Bathroom wall (to what height?)							
Balcony and utility							

Area	Tile type	Size	Brand and range	Code	Supply rate or allowance (₹ per sq ft)	Skirting included?	Who supplies
Staircase							
Terrace or parking							

Also fix: the tile adhesive or mortar to be used, the grout type and colour, and whether wastage is included in the quoted rate.

Section 3: Painting

Area	Putty (coats)	Primer	Paint type	Brand and range	Coats	Shade / colour code
Interior walls						
Ceilings						
Exterior walls						
Doors and grills (enamel)						

Section 4: Doors and windows

Item	Material	Size	Frame	Hardware (locks, hinges, handles)	Brand	Who supplies
Main door						
Bedroom doors						
Bathroom doors						
Windows	UPVC, aluminium or wood					
Window glass	Thickness and type					
Grills and safety						
Ventilators						

Section 5: Plumbing and sanitaryware

Item	Specification	Brand and model	Quantity	Who supplies
Water supply pipes	CPVC, UPVC or other; class			
Drainage pipes	Type and class			
Water closet (WC)	Wall-mounted or floor; flush type			
Wash basins				
Taps and mixers	Range			
Shower set				
Kitchen sink	Size and material			
Overhead tank	Capacity and type			
Sump	Capacity			
Water heater points	Number and location			

Section 6: Electrical

Item	Specification	Brand	Who supplies
Wires	Type (for example, FR or FRLS), sizes as per the electrician's circuit plan		
Conduits	Type and size		
Switches and sockets	Modular range		
Distribution board and MCBs			
Earthing	Type and number of pits		
Light fittings	Included or owner supplies?		
Fans	Included or owner supplies?		
AC points	Number and location		
Inverter wiring	Yes or no		

Also fix: the number of points per room. Use the Services tab in the workbook for the count.

Section 7: Kitchen

Item	Specification	Brand	Who supplies
Counter top	Material and thickness		
Counter length			
Cabinets	Material (for example, plywood grade, laminate)		
Chimney and hob points			
Water purifier point			

Section 8: Things contractors often leave out

Ask each contractor to confirm in writing whether these are included:

☐ Compound wall and gate
☐ Sump and septic tank, or connection to the sewer
☐ Rainwater harvesting
☐ Staircase railing and balcony railing
☐ Loft and shelves
☐ Electrical connection charges and meter
☐ Water connection charges
☐ Site clearing and debris removal
☐ Curing water and electricity during construction
☐ Scaffolding
☐ Final cleaning

Before you send this sheet

☐ Structure section confirmed by your engineer
☐ Every finish has a brand and code, or an allowance rate
☐ Every line says who supplies
☐ The same version is sent to every contractor, with a date
☐ Each contractor is asked to price line by line, not as one lump sum

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Bonus: Material Quality Checks on Site

Simple checks you can do yourself when materials arrive, before they are used.

> **Important:** these are field checks, not laboratory tests. They help you spot obvious problems early. Your engineer decides which checks apply, what the limits are, and whether laboratory tests are needed. Where this guide gives a figure, it is a commonly used field guide, not a standard. Follow your engineer's specification wherever it differs.

The routine for every delivery

1. **Count or weigh** what arrived, and compare it with the invoice.
2. **Check the markings:** brand, grade, ISI mark where the product carries one, batch and date.
3. **Do the quick checks** for that material, below.
4. **Store it properly** the same day.
5. **Record it** on the Material Receipt Log at the end of this guide, with a photo of the invoice and the markings.

If a check fails, **do not use the material**. Keep it aside, photograph it, and inform the supplier and your engineer the same day.

Cement

Check	How to do it	What is acceptable	If it fails
Grade and brand	Read the bag	Matches your specification sheet	Reject the delivery
ISI mark	Look for the mark and licence number on the bag	Present	Ask your engineer
Date of manufacture	Printed on the bag (week, month and year)	Fresh stock. A common site practice is to use cement within about three months of manufacture. Check with your engineer.	Ask your engineer before use
Lumps	Press the bag	Free-flowing, no hard lumps	Do not use lumped cement
Bag weight	Weigh a few bags at random	50 kg each, allowing a small tolerance	Raise with supplier
Bags intact	Look for tears and damp patches	Sealed and dry	Set aside

Storage: keep bags on a raised wooden platform or plastic sheet, away from walls, in a dry covered place. Stack them in limited heights (many sites limit stacks to about 10 bags). Use the oldest stock first.

TMT steel bars

Check	How to do it	What is acceptable	If it fails
Grade and brand	Read the markings rolled on the bar	Matches your specification (for example, Fe 500 or Fe 500D)	Reject
Test certificate	Ask the supplier for the manufacturer's test certificate	Provided for the batch	Ask your engineer
Diameter	Measure with a vernier caliper, or check by weight	Matches the size ordered	Raise with supplier
Weight per metre	Weigh a measured length. The workbook rule of thumb is diameter² ÷ 162 kg per metre.	Close to the rule-of-thumb figure	Ask your engineer
Rust	Look at the surface	Light surface rust that rubs off is usually acceptable. Flaking or pitted rust is not.	Ask your engineer
Cracks at bends	Look at any bars already bent	No cracks	Reject those bars

Worked weight check: a 12 mm bar weighs about $12^2 \div 162 = 0.89$ kg per metre. A 12 m bar should be about 10.7 kg. A bar much lighter than this may be under-size.

Storage: keep bars off the ground on supports, covered from rain, sorted by size.

Sand (M-sand or river sand)

Check	How to do it	What is acceptable	If it fails
Type	Look and ask	Matches your specification (concrete or plastering grade)	Reject
Silt content (field test)	See the jar test below	A commonly used field limit is about 8% by volume. Your engineer sets the limit.	Ask your engineer
Clay and dirt	Rub a handful between wet palms	Little or no stain left on the hands	Ask your engineer
Foreign matter	Look for roots, leaves and debris	Clean	Reject
Volume delivered	Measure the truck body or the heap	Matches the invoice	Raise with supplier
Test report (M-sand)	Ask the supplier	Provided	Ask your engineer

The jar test for silt: 1. Take a clear glass jar or measuring cylinder. 2. Fill it about halfway with sand. 3. Add clean water, with a pinch of salt, until it is about three-quarters full. 4. Shake well, then leave it to stand for about three hours. 5. Silt settles as a fine layer on top of the sand. Measure its height. 6. Silt % = silt layer height ÷ sand layer height × 100.

Coarse aggregate (jelly)

Check	How to do it	What is acceptable	If it fails
Size	Look and measure a few pieces	Matches the size ordered (for example, 20 mm)	Reject
Shape	Look at a handful	Mostly angular and cube-like. Few flat or long, thin pieces.	Ask your engineer
Cleanliness	Look for dust and mud coating	Clean	Wash before use, or reject
Volume	Measure the truck or heap	Matches the invoice	Raise with supplier

Red clay bricks

Check	How to do it	What is acceptable	If it fails
Size and shape	Measure a few at random	Uniform size, straight edges, sharp corners	Reject badly shaped lots
Colour	Look	Even colour across the lot	Ask your engineer
Sound	Strike two bricks together	A clear ringing sound	A dull sound suggests under-burnt bricks
Scratch	Scratch with a fingernail	No deep mark	Ask your engineer
Breakage	Count broken bricks in the delivery	Small number	Raise with supplier
Water absorption	Weigh a dry brick, soak it in water for 24 hours, weigh again	Your engineer sets the limit. Many specifications use about 20% for common bricks.	Ask your engineer
Count	Count stacks	Matches the invoice	Raise with supplier

AAC blocks

Check	How to do it	What is acceptable	If it fails
Size	Measure	Matches the order	Reject
Density and strength	Ask the supplier for the test certificate	Provided, matching your specification	Ask your engineer
Damage	Look for broken corners and cracks	Few damaged blocks	Raise with supplier

Check	How to do it	What is acceptable	If it fails
Jointing material	Check the adhesive or mortar supplied	As specified by the block maker	Ask your engineer

Water for construction

Check	How to do it	What is acceptable	If it fails
Appearance	Look and smell	Clear, no oil, no smell	Do not use
Taste	Taste a small amount from a clean source	Not salty	Ask your engineer
Source	Ask where it comes from	A known, clean source	Ask your engineer

Ready-mix concrete (if used)

Check	How to do it	What is acceptable	If it fails
Delivery ticket	Read it before unloading	Grade, quantity, and time of batching match your order	Do not accept
Time since batching	Compare the batching time with the arrival time	Within the time your engineer allows	Ask your engineer before unloading
Workability (slump)	Engineer or supervisor does a slump test	Within the specified range	Ask your engineer
Water added on site	Watch the unloading	No water added without your engineer's approval	Stop and ask
Test cubes	Cubes cast from the pour, labelled with the date	Cubes taken as your engineer requires	Insist they are taken

Concrete mixed on site

- ☐ The mix follows your engineer's specification, using measuring boxes, not shovels
- ☐ Water is measured, not added by guesswork
- ☐ Concrete is placed within the time your engineer allows after mixing
- ☐ It is compacted with a vibrator where your engineer requires one
- ☐ Test cubes are taken, labelled with the date, and sent for testing as your engineer requires
- ☐ Curing starts on time and continues for the period your engineer sets

Tiles

Check	How to do it	What is acceptable	If it fails
Brand, range and code	Read the boxes	Matches your specification sheet	Reject
Batch and shade	Read the batch and shade codes on every box	The same for each room	Separate the boxes. Do not mix shades in one room.

Check	How to do it	What is acceptable	If it fails
Size	Measure a few	Matches the box	Raise with supplier
Flatness	Place two tiles face to face	Little or no rocking	Ask your engineer
Chips and cracks	Open a few boxes	Few damaged pieces	Raise with supplier
Quantity	Count boxes	Matches the invoice	Raise with supplier

Paint

☐ Brand, range and shade code match your specification sheet
☐ Tins are sealed
☐ Batch number and date are printed on each tin. Record them.
☐ Primer and putty are the products specified

Electrical wires and fittings

☐ Brand and type match your specification (for example, FR or FRLS)
☐ The wire size (sq mm) printed on the insulation matches your electrician's circuit plan
☐ The ISI mark is present where the product carries one
☐ The coil length printed on the pack matches what you were charged for
☐ Switches, sockets and MCBs match the brand and range you chose

Plumbing pipes and fittings

☐ Pipe type and class or pressure rating are printed on the pipe and match your specification
☐ The ISI mark is present where the product carries one
☐ Fittings are from the same brand as the pipes, unless your plumber advises otherwise
☐ Sanitaryware model numbers match your specification sheet

Material Receipt Log

Print one page per month. Record every delivery.

Date	Material	Supplier	Invoice number	Quantity on invoice	Quantity checked	Markings (grade, batch, date)	Checks passed?	Photo taken?	Received by

Construction Calculator Pro. General guidance only. Your engineer, contractor, lawyer and local authority decide what applies to your project.

Bonus: Contractor Hiring Kit

How to find, check and choose a contractor before you sign.

The process at a glance

1. Build a shortlist of four to six contractors.
2. Call each one with the same questions.
3. Visit at least two of their sites: one finished and one in progress.
4. Speak to two past clients for each contractor.
5. Send the same specification sheet to three contractors and ask for itemised quotes.
6. Compare the quotes line by line on the Bids tab.
7. Score each contractor on the scoring sheet.
8. Check the agreement before you sign or pay anything.

Step 1: Build a shortlist

Good sources: - Owners of houses near you that were built in the last three years. Ask who built them, and whether they would hire the same contractor again. - Your architect or structural engineer. - Building material dealers, who see which contractors pay on time and buy proper material.

Note each contractor's name, firm name, phone, the area they work in, and who referred them.

Step 2: The first call

Ask every contractor the same questions, and write the answers down.

1. How many houses have you completed in the last three years? Can I see two of them?
2. Do you have a site running now that I can visit?
3. How many sites do you run at the same time?
4. Who will supervise my site day to day? Will that person be on site every day?
5. Do you use your own workers, or subcontract? Which trades do you subcontract?
6. Do you quote item by item against my specification, or as a rate per sq ft?
7. If per sq ft, what exactly is included, and what is not?
8. How do you handle changes that come up during the work?
9. What payment stages do you usually ask for?
10. What do you do if the work is delayed?
11. What do you guarantee after handover, and for how long?
12. Do you have a GST registration? Will you give proper invoices?

Step 3: Visit their sites

At a finished house:

☐ Walls straight and plaster even, with no visible cracks
☐ Tiles even and not hollow when tapped
☐ Doors and windows close properly
☐ No damp patches on ceilings or below windows
☐ Bathroom floors slope towards the drain

At a site in progress:

☐ Materials stored properly (cement off the ground and covered, steel off the ground)
☐ The site is reasonably tidy and safe
☐ A supervisor is present
☐ Curing is being done
☐ Workers have basic safety equipment

Step 4: Speak to past clients

Ask each past client: 1. Did the house cost about what was quoted? If not, why not? 2. Was it finished on time? If not, how late, and why? 3. How were changes handled? Were they priced before the work? 4. Did the contractor ask for money ahead of the work? 5. Were there problems after handover? How quickly were they fixed? 6. Would you hire this contractor again?

The last question matters most. A hesitant answer is a warning sign.

Step 5: Ask for itemised quotes

Send the same Specification and Brand Sheet to three contractors, with the same date and deadline. Ask each to: - Price line by line, against your specification - State what is excluded - State the payment stages they want - State the expected duration

A single rate per sq ft is hard to compare and hard to check. If a contractor will only quote per sq ft, ask for a written list of what is included in that rate.

Step 6: Compare the quotes

Use the Bids tab in the workbook. For each item: - Check that each contractor priced the same specification. - Check that the quantities are close to your own from the workbook. - Look for items left out of one quote. A low total can mean missing items. - Look for very low rates. Ask how the contractor will achieve them.

Do not choose on price alone. The cheapest quote often has the most exclusions, which come back as extras later.

Step 7: Scoring sheet

Score each contractor from 1 (poor) to 5 (excellent). Multiply each score by the weight, then add.

Factor	Weight	Contractor A	Contractor B	Contractor C
Quality of finished houses seen	3			
Past clients' feedback	3			
Site supervision and organisation	2			
Clarity and completeness of quote	2			
Price, after adding back missing items	2			
Payment terms (pay after work, not ahead)	2			
Communication and responsiveness	1			
Total (out of 75)				

These weights are a starting point. Change them to match what matters to you.

Warning signs

- Refuses to let you visit a past or running site
- Pushes a quick decision, or a discount that expires today
- Will not give an itemised quote, or will not say what is excluded
- Asks for a large advance before any work starts
- Asks for cash without receipts
- Has no supervisor, or the owner is never on site
- Past clients are hard to reach, or hesitant
- Quote is far below the others, with no clear explanation

Step 8: Before you sign

☐ The agreement lists the scope, and attaches the specification sheet and drawings
☐ Payment stages are linked to completed work, not to dates
☐ The change order process is written down (see the Change Order Control bonus)
☐ The duration, and what happens if work is delayed, is written down
☐ What is guaranteed after handover, and for how long, is written down
☐ Any amount held back until snags are fixed is agreed
☐ Your lawyer has read the agreement

This kit is general guidance. It is not legal advice.

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Bonus: Snag List and Handover Check

Find defects before the final payment, while the contractor still has a reason to fix them.

When to do it

- **First walk-through:** when the contractor says the work is finished, before the final payment.
- **Second walk-through:** after the contractor says the snags are fixed. Check each one again.
- **After moving in:** keep a list of anything that appears in the first few months, for the guarantee period agreed in your contract.

Agree in the contract, before work starts, how much of the final payment is held back until the snags are fixed. Hold that amount until the second walk-through is complete.

What to bring

☐ A spirit level (600 mm or longer)
☐ A measuring tape
☐ A torch
☐ A coin or the handle of a screwdriver, to tap tiles
☐ A small ball or a marble, to check floor slopes
☐ A bucket of water
☐ A phone charger or a small lamp, to test sockets
☐ Masking tape and a marker, to label each snag on the wall
☐ This checklist, and your phone for photos

Take a photo of every snag, and write the number on the masking tape label in the photo.

Room-by-room checks

Walls and ceilings

☐ No cracks in plaster. Note hairline cracks too.
☐ Walls look straight when you look along them with the torch held close to the surface
☐ No hollow sound when you tap the plaster
☐ Corners and edges are straight and even
☐ No damp patches, especially below windows, on ceilings under bathrooms, and on the top floor ceiling

Paint

☐ Even colour, with no patches, drips or brush marks

☐ Paint does not come off when rubbed lightly
☐ Edges are clean against ceilings, switches, doors and tiles
☐ No paint splashes on floors, glass, switches or fittings

Floors and tiles

☐ No hollow sound when you tap each tile, especially the corners. A hollow sound means the tile may crack or come loose later.
☐ Tiles are level with each other at the joints. Run your hand across them.
☐ Grout is complete, even, and the agreed colour
☐ No chipped or cracked tiles
☐ Skirting is fixed and even
☐ Shades match within each room

Doors

☐ Opens and closes without rubbing or sticking
☐ Stays open where you leave it, without swinging
☐ Locks and latches work with each key
☐ Even gap around the door when closed
☐ Stoppers and handles are fixed firmly
☐ Bathroom doors are protected at the bottom against water, as specified

Windows

☐ Opens, slides and closes smoothly
☐ Locks work
☐ Glass has no cracks or scratches
☐ Seals and gaskets are complete
☐ **Water test:** pour water on the outside of the window frame. No water should come inside.
☐ Grills and mosquito mesh are fixed

Electrical

☐ Every switch works its light or fan
☐ Every socket works. Test each one with the charger or lamp.
☐ Switch plates are straight and fixed firmly
☐ The distribution board is labelled, circuit by circuit
☐ MCBs switch off each circuit correctly
☐ Earthing has been tested by the electrician. Ask for the reading.
☐ Fans and lights supplied by the contractor are fixed and working
☐ AC, water heater and inverter points are in the agreed places

Plumbing

☐ Every tap gives a steady flow of water
☐ Hot and cold are on the correct side
☐ No leaks under sinks and basins. Check with the torch after running water for a few minutes.
☐ Each WC flushes properly and refills, without running continuously
☐ Drains clear quickly, without gurgling or backing up
☐ No smell from drains
☐ Overhead tank fills and stops correctly

Bathrooms

☐ **Slope test:** pour a bucket of water on the floor. It should run to the drain and leave no standing water.
☐ Water does not flow out of the bathroom door
☐ No damp on the walls outside the bathroom, or on the ceiling below
☐ Fittings (taps, shower, towel rails) are fixed firmly
☐ Wall tiles are not hollow and joints are sealed

Kitchen

☐ Counter is level. Check with the spirit level.
☐ Sink drains quickly and does not leak underneath
☐ Cabinet doors and drawers open and close properly
☐ Chimney, hob and appliance points are where agreed
☐ Wall tiles behind the counter are complete and sealed

Staircase and railings

☐ Steps are even in height
☐ Railings are firm and fixed at every support
☐ No sharp edges on railings or grills

Terrace and roof

☐ Waterproofing is complete, as specified
☐ **Ponding test:** if your engineer agrees, block the drains and keep water on the terrace for the period your engineer advises. Check the ceiling below for damp.
☐ Drain outlets are clear, with grates fitted
☐ Water flows to the outlets without pooling
☐ Parapet walls have no cracks

Exterior

☐ Paint is even and complete
☐ No cracks in external plaster

☐ Rainwater pipes are fixed and lead to drains or the rainwater harvesting pit
☐ Sump and septic tank covers are fitted and safe
☐ Compound wall and gate are complete, if included
☐ Site is cleared of debris

Documents to collect at handover

☐ Completion or occupancy certificate, as required by your local authority
☐ Approved drawings and as-built drawings, including plumbing and electrical routes
☐ Structural drawings
☐ Warranty cards and invoices for fittings, pumps, water heaters and other equipment
☐ Test reports (concrete cubes, materials) kept during construction
☐ Electricity and water connection documents
☐ All keys, labelled
☐ Contact details for the plumber and electrician who did the work

Snag record

No.	Room	Location	Description	Photo?	Reported on	Fixed on	Checked by
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							

Handover sign-off

All snags fixed and checked on	
Documents received (Yes/No)	
Final payment made on	

Amount held back, if any (₹)	
Owner (name and signature)	
Contractor (name and signature)	

This checklist is general guidance. Your engineer decides which tests apply, and how long a ponding test should run.

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